Yorkshire & Humber Region Crime Analysis Report

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Introduction

The Yorkshire and Humber region, encompassing the largest historic county of England, is comprised of a rich tapestry of cultural heritage with stunning landscapes and vibrant cities. To manage and respond to crimes in this region effectively, the Yorkshire and Humber Regional Organized Crime Unit (YHROCU) requires thorough data driven insights. From the vast data crime that has been accumulated through the years, we created an interactive dashboard to understand the trends and patterns from these datasets. Our dashboard contains data from years 2022 and 2023 spanning across three main sections – Street Crimes, Stop and Search and Crime Outcomes. We inspected and visualized important factors like most common crime types, Cities that are affected the most and the rate of crime over the months. This report helps analyze prominent insights, patterns and trends over the course of the two years in the Yorkshire and Humber region.

Data Source & Extraction

The datasets used to visualize our dashboard was obtained from the UK official police data website: https://data.police.uk/data/ that includes public datasets containing information on street crime, stop and search statistics and police investigation outcomes. We downloaded a total of 264 datasets which were spread across 24 months. They contain data from areas like West Yorkshire, South Yorkshire, North Yorkshire and Humberside which includes all the Steet crime, stop and searches and Crime outcomes from the beginning of 2022 to the end of 2023.

Then we imported the datasets into SQL Server Management System using the import flat file option. We tried to use the import wizard but it wasn't working properly since we couldn't properly change data types during the import. The 264 datasets had to be imported one by one since the Bulk loading command wasn't working as we had hoped and we only found the way to insert all the dataset easily with command interface only after importing it one by one. We imported the datasets into multiple tables, then categorized and inserted them into three main tables named StreetCrime, StopAndSearch and CrimeOutcomes. Finally, we dropped the other tables that wasn't needed anymore.

Basic code for insertion of table contents to the main table:

```
INSERT INTO [Maintable] (Columns in the table)

SELECT [Columns from the other table]

FROM [Othertable];
```

Basic code to drop table:

```
DROP TABLE [Tablename];
```

Data Cleaning

Then we had to clean the data in our three tables so that there are no null values and duplicates.

We used the following code to remove duplicates;

```
WITH CTE_Duplicates AS (
   SELECT *,
        ROW_NUMBER() OVER (
            PARTITION BY Crime_ID, Month, Reported_by, Falls_within, Longitude, Latitude,
Location,LSOA_code, LSOA_name,Crime_type,Last_outcome_category,Context
            ORDER BY Crime_ID) AS RowNum
   FROM dbo.StreetCrime
)
DELETE FROM CTE_Duplicates
WHERE RowNum > 1;
```

- CTE-common table expression; it is a temporary result set, which is named and derived from a simple query. It's defined within the scope of a single SELECT, INSERT, UPDATE, or DELETE statement. This query removes duplicate rows by keeping only the first occurrence of each unique row. The ROW_NUMBER() function assigns a unique row number to each duplicate row in partitions based on the specified columns. Rows with a row number greater than 1 are considered duplicates and are deleted, leaving only one unique row for each set of duplicate data.

When removing null values, we set the null values to ‘unknown’ for the columns with the data type nvarchar and ‘0’ to the columns with the data type – float (i.e. Latitude and Longitude).

The simple code is as below;

```
UPDATE dbo.StreetCrime
SET Crime_ID = 'Unknown'
WHERE Crime_ID IS NULL;
```

After cleaning the data, we backed up our database to a bak file. Then we opened Power BI and imported the three tables by connecting our SQL database to Power BI.

User Interface & Functionality

The dashboard has a user-friendly interface that easy for navigation. The first page which is the home page has a dashboard design where the user can choose which page or section they want to see. By clicking on an option, it takes the user to their chosen page. On the top of the page there is a filter icon that you can click on where you get a pop-up filter pane where you could use filters to choose the year and the LSOA name for easier access to data in a specific region or a specific year. You can reset everything to the original state by clicking on the reset icon in the filter pane. You can navigate through the pages by clicking on the page navigation buttons on top of each page.

We used DAX for making almost all the cards in dashboard. We used it to take percentages of the most common outcomes. We have applied filters for the maps, where the latitude and Longitude will only display values that are not zero. By that, the null values won't affect the map visual and you can only see the UK map, not the map of the whole world. It is through the combination of these navigation options, filtering capabilities and detailed view functionality, that the user will be able to cover and analyze the data presented on the dashboard effectively and efficiently.

DAX codes used for the cards;

```
MostCommonCrimePercentage =
VAR OutcomeSummary =
  SUMMARIZE(
    StreetCrime,
    StreetCrime[Crime_type],
    "OutcomeCount",
    COUNT(StreetCrime[Crime_type])
  )
VAR TopOutcome =
  TOPN(
    1,                        -- Return the top 1 result
    OutcomeSummary,           -- Use the summarized table
    [OutcomeCount], DESC      -- Sort by the count in descending order
```

```
  )
VAR TopOutcomeCount = MAXX(TopOutcome, [OutcomeCount])  -- Get the count of the top outcome
VAR TotalOutcomes = COUNTROWS(StreetCrime)            -- Get the total number of outcomes
RETURN
  DIVIDE(TopOutcomeCount, TotalOutcomes, 0)          -- Return a decimal value (e.g., 0.3456)
```

- The SUMMARIZE function creates a summary table and groups street crime by crime type and counts the number of time each occurs using the COUNT function. The TOPN function returns the row from the summary outcome table where the count is the highest. The topoutcomecount variable uses the MAXX function to get the occurrences of the most frequent crime. The COUNTROWS function is used to count the total number of rows. Finally the RETURN statement used to calculate the percentage of the most common crime type.

Trends

Street Crime

As per our visualizations in page 2, we can initially observe that the total count of crime is reported as 1.34M, indicating a significant volume of criminal activity.

Violence & Sexual offences account for 39.5% of all crimes, making it the most frequent type. This high percentage is concerning regarding the public safety and suggests a need for targeted interventions in this area.

The most common outcome for reported crimes is “Unable to Prosecute” at 38.6%. This high rate of unprosecuted crimes could potentially undermine public confidence in the justice system.

The count of crime by crime type bar chart shows a distribution of various crime types with violence and sexual offences having the largest count – 529710. It is followed by public order, criminal damages and anti-social behavior respectively. The graph shows clear dominance by a few major categories, especially by violence & sexual offences with lesser crimes like drugs, burglary and bicycle theft contribute to smaller portions.

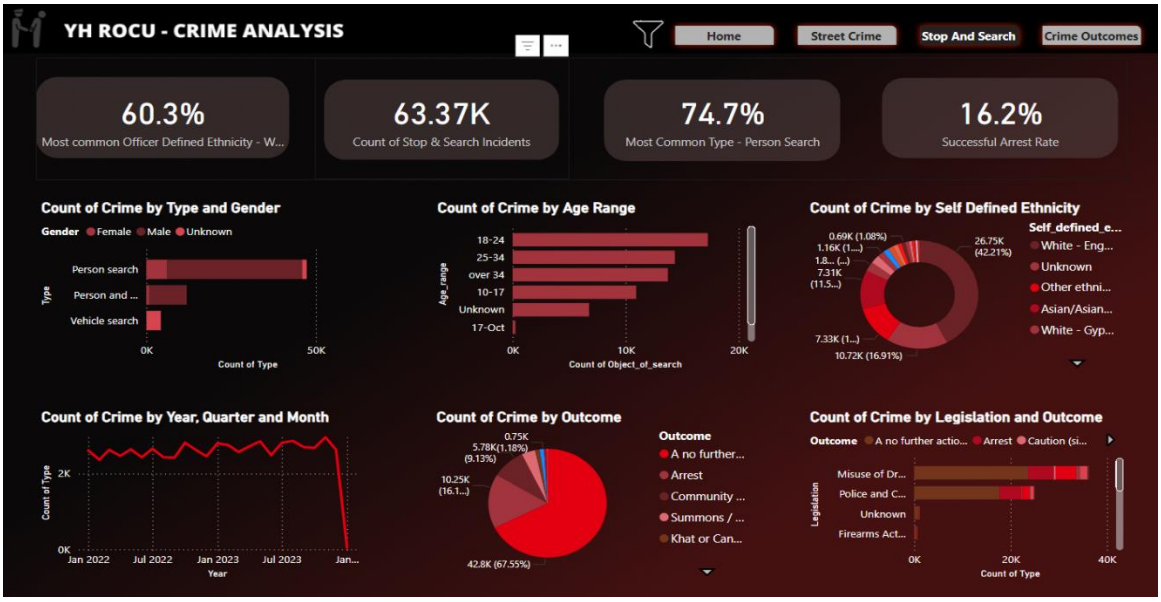
The count of crime by month bar chart shows a steady crime along the months with only slight increases in crime during the summer months, especially July and noticeable decrease in crime during December. These slight changes are possibly because of seasonal fluctuations.

The line chart representing the count of crime by year and crime type shows the crime trends spread across the two years. The graph shows relatively stable patterns across the years for the crime types with some seasonal fluctuations consistent across the two years. The most common types like violence and sexual offences continue to show higher trends, while the less common types are not that frequent but they remain constant.

Th pie chart which represents the count of crime by last outcome category provides us insights which suggests that a large number of cases fail to reach resolution through legal action. This is because the most prominent outcomes are “Unable to prosecute” with a output of 38.56%, 35.18% being “Investigation complete- no suspect identified” and 8.79% results being unknown.

The count of crime by LSOA code and crime type area chart visualizes the distribution of crime across various regions. The areas with high crime frequencies may likely be regions with high population or popular crime hotspots for certain crimes.

Stop And Search



The visualizations in page 3 focuses on stop and search incidents and provides more detailed demographic information about crime in the area.

The total number of stop and search incidents is 63.37k. This large amount indicate that this law enforcement tactic is commonly used in this region. The most common officer defined ethnicity of stop and search individuals are white, accounting to 60.3%. The most common type is person search, which is about 74.4%. This could mean that the police are targeting suspicious people and their behavior instead of vehicles or objects. Only 16.2% of these stop and search incidents end in arrests.

The bar chart that represents the count of crime by type and gender shows that males (33773) are significantly more likely to be subjected to be stopped, especially for person searches than females.

The count of crime by age range bar chart shows that a large number of stop and searches occur in the 18 – 24 and 25 – 34 age groups. These age ranges typically have a higher proportion of offenders while fewer stops occur for minors (10 -17) and older adults.

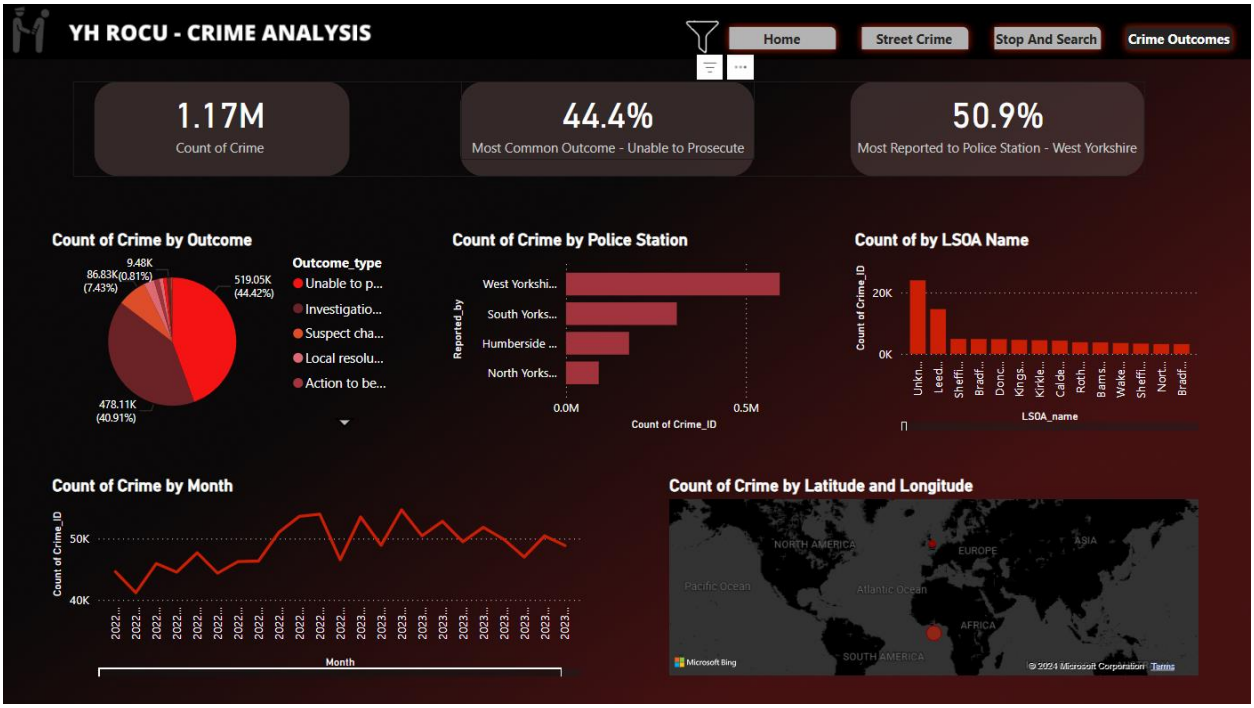
The donut chart shows that white – English/Welsh/Scottish/British ethnic group is the largest category at 42.21% followed by unknown at 16.91%. Notable proportions are from Asian and other ethnic backgrounds.

The line chart representing count of crime by year, quarter and month shows fluctuations over time but there are no major increases or decreases. The sharp drop at the end indicates recent data rather than a decline of crime.

The pie chart for crime outcomes show that the majority of crimes result in no further action disposal (67.55%), followed by only 16.1% leading to an arrest. There are only smaller percentages for outcomes like community resolution and summons/ charged by post.

The Legislation is visualized by a stacked bar chart that suggests Misuse of drugs and police criminal evidences are the most common legislative grounds for stop and searches.

Crime Outcomes



Page 4 of the dashboard provides an overview of crime outcomes and geographical distribution.

The total count of crime is reported as 1.17M and the most common output is inability to prosecute at 44.4%. 50.9% of crimes are reported to West Yorkshire Police station, indicating a high concentration of crime in this area.

The pie chart representing count of crime by outcome shows 44.42% as unable to prosecute, 40.91% being under investigation and only 7.43% resulting in the suspect being charged in some way. This indicates a large proportion of cases do not result in identifying or prosecuting suspects.

The count of crime by police station bar chart shows that West Yorkshire receives the highest number of crime reports, followed by South Yorkshire, Humberside and North Yorkshire respectively. This could be due to crime distribution, population differences or law efficiency in the areas.

The bar chart representing the count of crime by LSOA name shows crime across various lower layer super output areas with significant variations between the areas. We can notice that crimes are highly concentrated in areas like Leeds and Sheffield which could be crime hotspot areas.

The count of crime by month chart shows fluctuations overtime, with an overall slight upward trend visible. Crime seems to peak during certain times of the year like summer months and go slightly downwards during the end of the year due to seasonal factors.

Conclusion

This Yorkshire and Humber crime analysis dashboard provides a critical look into the trend of crime, demographic component, and policing activities across the years 2022 and 2023. Key observations from this analysis include but are not limited to the following:

1. Within the region, violent and sexual offenses take up about 40% of the crimes reported. This gives a direct indication that specific intervention methods and preventative measures are very necessary in this domain.

2. The inability to prosecute a high percentage of cases-38.6 percent to 44.4 percent-is a cause for concern that could erode public confidence in the system. This points to a need to develop further techniques and resources for investigation.

3. Stop and search practices, though conducted frequently, lead to comparatively few arrests at a rate of 16.2%. This raises questions about the effectiveness and possible biases in such operations, especially given the demographic breakdowns observed.

4. There is distinct geographic concentration of crime, as more than half of the crimes reported took place in West Yorkshire. This can be useful in informing resource allocation across the region.

5. Seasonal trends in crimes that increase during the summer months give a wealth of information for proactive policing strategies.

6. The demographic data on stop and searches, particularly regarding age, gender, and ethnicity, warrants further examination to ensure fair and effective policing practices.

This will be a powerful tool for law enforcement agencies, policymakers, and community leaders. Guided by data insights from the dashboard, stakeholders will have more focused and effective ways of reducing crimes, improving public safety, and enhancing the effectiveness of the criminal justice system in Yorkshire and Humber.

As such, continued updating and the analysis of this information will be important in monitoring the effects of any changes instituted through time, along with the identification of emerging trends. Also, its integration with other socio-economic indicators could provide a better understanding of the driving forces behind crime within the region.

Resources

Tips for IT Pros, SQL Server Remove Duplicate Rows with CTE - https://www.youtube.com/watch?v=6ycNqvaMCzs

Jason Davidson, Creating a Pop-Out Navigation and Filter Bar Using Power BI Bookmarks. 4 Ideas for Sections to Add - https://www.youtube.com/watch?v=SCI9iKl4vjU&t=1096s

SQLBI, Measures vs. calculated columns in DAX and Power BI - https://www.youtube.com/watch?v=ePPi1LLX0sA&list=LL&index=1&t=45s

Kevin Stratvert, How to use Power BI DAX – Tutorial - https://www.youtube.com/watch?v=waG_JhBgUpM&t=341s

Microsoft Learn, https://learn.microsoft.com/en-us/dax/topn-function-dax